

SELF-STUDY REPORT

1. Runtime JIT Overhead

Just-In-Time (JIT) compilation converts Intermediate Language (IL) code into machine code at runtime, which introduces initial startup latency and memory overhead.

This compilation delay can impact cold-start performance, particularly in serverless applications and lightweight microservices.

Understanding JIT dynamics helps developers optimize execution paths and resource allocation during application boot.

2. Techniques to Reduce JIT Overhead

Ahead-Of-Time (AOT) compilation mitigates latency by compiling IL directly into native machine code prior to execution.

Additionally, Tiered Compilation balances startup speed and runtime throughput by progressively optimizing frequently executed code blocks.

Features like ReadyToRun (R2R) images further accelerate boot times by storing pre-compiled native code alongside assemblies.

3. Hello World Setup via VS Code

Setting up C# development in VS Code requires installing the .NET SDK along with the C# Dev Kit extension.

Creating a new console application is performed quickly through the command line using the command `dotnet new console`.

Running `dotnet run` compiles and executes the program seamlessly directly within VS Code's integrated terminal.